

Pyelonephritis Caused by *Pseudomonas aeruginosa*: From Treatable Infection to Therapeutic Dead-End, Rescued by an Investigational Antibiotic

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Pseudomonas aeruginosa is a major Gram-negative pathogen causing diverse infections, including complicated urinary tract infections (cUTIs). Although such cUTIs are usually treatable with standard antipseudomonal β -lactams, the rise of carbapenem-resistant *P. aeruginosa*, particularly those producing New Delhi metallo- β -lactamase (NDM), has made therapeutic options extremely limited. We report a 58-year-old man with diabetes, hypertension, and ischemic heart disease who presented with pyelonephritis in the setting of a long-retained ureteric stent. Urine culture grew *P. aeruginosa* resistant to meropenem, and molecular testing confirmed the presence of NDM. The whole-genome sequencing revealed that the isolate belonged to high-risk sequence type 308 clone carrying NDM-1 and multiple additional resistance determinants. Given the lack of effective alternative antibiotics and the patient's deteriorating clinical course with underlying renal impairment precluding the use of colistin, investigational drug cefepime/zidebactam (WCK 5222) was initiated under compassionate ground. The patient underwent source control with percutaneous nephrostomy and stone clearance. He received 37 days of cefepime/zidebactam, during which he showed rapid clinical improvement, resolution of pyelonephritis, and sterile urine cultures. This case underscores how infections such as cUTI or pyelonephritis, usually considered manageable, can become untreatable in NDM-endemic settings and highlights cefepime/zidebactam as a promising therapeutic option for highly resistant *P. aeruginosa*.

Keywords: pyelonephritis, cefepime/zidebactam, India, WCK 5222, NDM, *P. aeruginosa*, compassionate-use

Introduction

Pseudomonas aeruginosa is one of the clinically important Gram-negative pathogens responsible for a wide spectrum of infections, including pneumonia, skin and soft-tissue infections, bloodstream infections, and complicated urinary tract infections (cUTIs). Among these, cUTIs and skin and soft-tissue infections are generally considered less severe than pneumonia or bacteremia and are typically manageable with appropriate intravenous antipseudomonal antibiotics.^{1,2}

In recent years, however, the rise of difficult-to-treat *P. aeruginosa* has created a significant therapeutic challenge.

Further, the emergence of carbapenemase-producing strains, particularly those harboring New Delhi metallo- β -lactamase (NDM), has dramatically restricted available treatment options.^{3,4} Under such circumstances, even infections usually regarded as straightforward to manage, such as UTIs, may evolve into situations where effective therapy is severely limited or altogether unavailable.

Here, we describe a case of pyelonephritis caused by NDM-producing *P. aeruginosa* in India, a region where NDM is endemic. The case illustrates how an otherwise manageable infection can represent a therapeutic dead end.⁵ The patient was ultimately treated successfully with the investigational β -lactam/

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β -lactam enhancer combination cefepime/zidebactam, underscoring the urgent need for novel antibiotics to manage highly resistant Gram-negative infections in endemic settings.

Case Presentation

A 58-year-old man with diabetes mellitus, hypertension, and ischemic heart disease underwent left ureteric Double-J (DJ) stent placement for renal calculi in September 2022. He defaulted on follow-up, leaving the stent *in situ* for more than a year. In December 2023, he developed hematuria and fever, which was managed empirically at another hospital (details not available). In January 2024, he was again hospitalized elsewhere with acute coronary syndrome and started showing persistent fever, leukocytosis, pyuria, and elevated serum creatinine (2.2 mg/dL). Imaging revealed left hydronephrosis and pyelonephritis, and the DJ stent was replaced, and the patient was referred to our center for further follow-up.

On admission, he was hemodynamically stable with a serum creatinine of 2.64 mg/dL and creatinine clearance of about 30.2 mL/min. Ultrasonography and contrast-enhanced computed tomography (CT) demonstrated a bulky left kidney with multiple renal calculi (largest 25 mm), mild hydronephrosis, and pyonephrosis (Fig. 1a, b). Pending culture and susceptibility results, empirical meropenem therapy (1 g BID) was initiated. Escalation to colistin was avoided at this stage, given the patient's renal impairment and the risk of nephrotoxicity.

Urine culture subsequently yielded *P. aeruginosa* resistant to meropenem, and Carba-R GeneXpert (off-label use for *P. aeruginosa*) confirmed the presence of the NDM carbapenemase. Therapeutic options were extremely limited: ceftazidime/avibactam plus aztreonam has uncertain efficacy against

metallo- β -lactamase (MBL)-producing *P. aeruginosa*;⁶ cefiderocol was not available in India (subsequent testing later demonstrated intermediate susceptibility); and colistin was contraindicated due to renal dysfunction.⁷

Given these constraints and the patient's clinical deterioration, the process to procure cefepime/zidebactam (WCK 5222) under compassionate ground was initiated with meropenem (1 g BID) continuing till then. Broth microdilution testing as per Clinical and Laboratory Standards Institute (CLSI) guidelines demonstrated a cefepime/zidebactam minimum inhibitory concentration (MIC) of 8/8 mg/L, which is well below the CLSI-endorsed investigational susceptibility breakpoint of $\leq 64/64$ mg/L for *P. aeruginosa*, thereby confirming susceptibility (Table 1).⁸ Therapy was initiated on March 4, 2024, with a renal dose adjustment of 1 g/0.5 g every 8 hours with a 60-minute infusion (corresponding to 50% of the standard dose of 2 g/1 g every 8 hours recommended for patients with creatinine clearance >60 mL/min). The following day, ultrasound-guided percutaneous nephrostomy drained purulent material again showed presence of *P. aeruginosa* with a similar resistance profile.

Over the next nine days, the patient showed marked improvement, with defervescence, resolution of flank pain, and a C-reactive protein declining from 136 mg/L to 16.5 mg/L. On day 10, he underwent retrograde intrarenal surgery, percutaneous nephrolithotomy, and DJ stent replacement (Fig. 1c).

On day 17, pruritus developed, leading to cefepime/zidebactam dose reduction (500/250 mg every 8 hours) and antihistamine therapy. By day 19, urine culture was sterile. The DJ stent was removed on day 31, and after resolution of pruritus, the cefepime/zidebactam dose was increased to 1/0.5 g every 8 hours from days 32 to 37. A urine culture on day 33

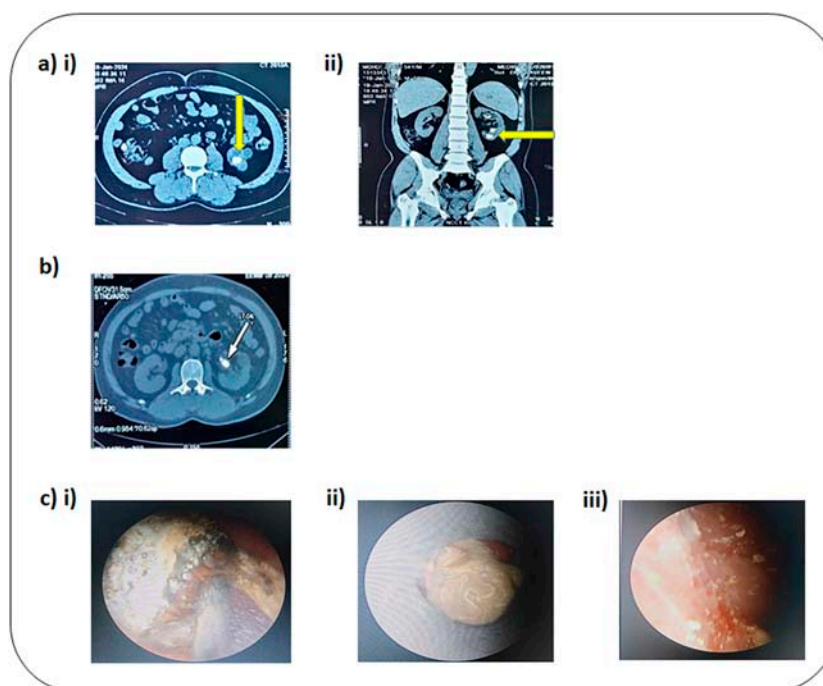


FIG. 1. Images showing renal calculi and acute pyelonephritis with pyonephrosis. (a) CT of the Kidneys, Ureters, and Bladder (KUB) with yellow arrows indicating the renal calculi. (i) CT KUB (axial view) showing left renal calculi; (ii) CT KUB (coronal plane) showing left renal calculi with moderate hydronephrosis with DJ stent in upper calyx; (b) CT KUB (axial plane) showing left acute pyelonephritis with pyonephrosis; (c) (i) stone fragmented using lithoclast, (ii) stone in calyx of kidney, (iii) pelvicalyceal system of kidney after stone removal.

TABLE 1. ANTIBIOTIC SUSCEPTIBILITY OF THE *P. AERUGINOSA* ISOLATE RECOVERED FROM THE PATIENT'S URINE SAMPLE

Antibiotics	MIC (mg/L)	Susceptibility (CLSI criteria)
Cefepime/zidebactam (1:1)	8	S ^a
Cefiderocol	8	I
Aztreonam/avibactam	32	R ^b
Imipenem	>128	R
Meropenem	>128	R
Ceftazidime/avibactam	>128	R
Piperacillin/tazobactam	>128	R
Ceftolozane/tazobactam	>128	R
Imipenem/relebactam	>128	R
Colistin	0.5	I
Amikacin	>128	R
Levofloxacin	128	R

^aFor cefepime/zidebactam, CLSI endorsed investigational susceptible breakpoint of ≤64/64 mg/L was used.

^bBased on aztreonam CLSI criteria.

S: susceptible, R: resistant, I: intermediate; interpretation as per CLSI criteria.

remained sterile, and follow-up imaging confirmed resolution of infection and clearance of calculi. Figure 2 explains a timeline of key clinical events, treatment progression, microbiological findings, and outcome.

The isolate was subjected to whole genome sequencing, which revealed that the isolate belonged to sequence type 308, an international high-risk clone and harbored NDM-1 variant,^{9,10} *bla*_{PAC-1}, and *bla*_{OXA-10}, with resistance mutations in genes encoding the following proteins: *gyrA*, *parC* and *parE* (linked to fluoroquinolone resistance), aminoglycoside acetyltransferase (*aac*), aminoglycoside adenyltransferase (*aadA1*) (Supplementary Table S1).

Discussion

Infections caused by wild-type *P. aeruginosa* are typically treated with β-lactams such as ceftazidime, cefepime, or

piperacillin/tazobactam, with carbapenems such as meropenem reserved for patients with risk factors; fluoroquinolones and aminoglycosides serve as additional alternatives depending on the clinical scenario. Over time, difficult-to-treat strains have emerged that are not amenable to first-line and safer agents (β-lactams and fluoroquinolones). Such infections may still respond to newer β-lactam/β-lactamase inhibitor combinations, including ceftolozane/tazobactam, ceftazidime/avibactam, imipenem/relebactam, or cefiderocol.^{11,12} However, when the pathogen harbors NDM, which is commonly encountered in India, therapeutic options become severely limited. Although cefiderocol can be active against MBL-producing *P. aeruginosa*, studies have shown limited susceptibility among Indian NDM isolates,⁴ and in the present case the isolate was resistant to cefiderocol. The use of ceftazidime/avibactam plus aztreonam, while effective for MBL-producing Enterobacterales, is not recommended for MBL *Pseudomonas* infections due to vulnerability of aztreonam to hyper-efflux which cannot be tackled by avibactam.^{6,13,14} This is evident from a report showing MICs of aztreonam/avibactam against most MBL-producing *P. aeruginosa* being above 8 mg/L (aztreonam susceptible breakpoint for *P. aeruginosa*: CLSI criteria).¹⁵ Colistin remains a potential option for urinary infections caused by MBL *P. aeruginosa* infection, but in this patient, renal impairment made it unsafe, imposing a true therapeutic dead end. In this scenario, we selected cefepime/zidebactam, supported by translational *in vivo* studies showing potent killing of MBL *P. aeruginosa* at its humanized regimens,^{16–18} and then two published cases of NDM *P. aeruginosa* infections (life-threatening intra-abdominal and necrotizing soft-tissue infections) treated with cefepime/zidebactam.^{19,20}

Phase 1 studies have demonstrated safety of cefepime/zidebactam, with cefepime as a known cephalosporin antibiotic and zidebactam belonging to the diazabicyclooctane pharmacophore with novel β-lactam enhancer mechanism and a favorable safety profile.²¹ Thus the benefit-risk assessment clearly favored the use of cefepime/zidebactam. The patient responded well, with resolution of chronic pyelonephritis following standard source control and cefepime/zidebactam treatment.

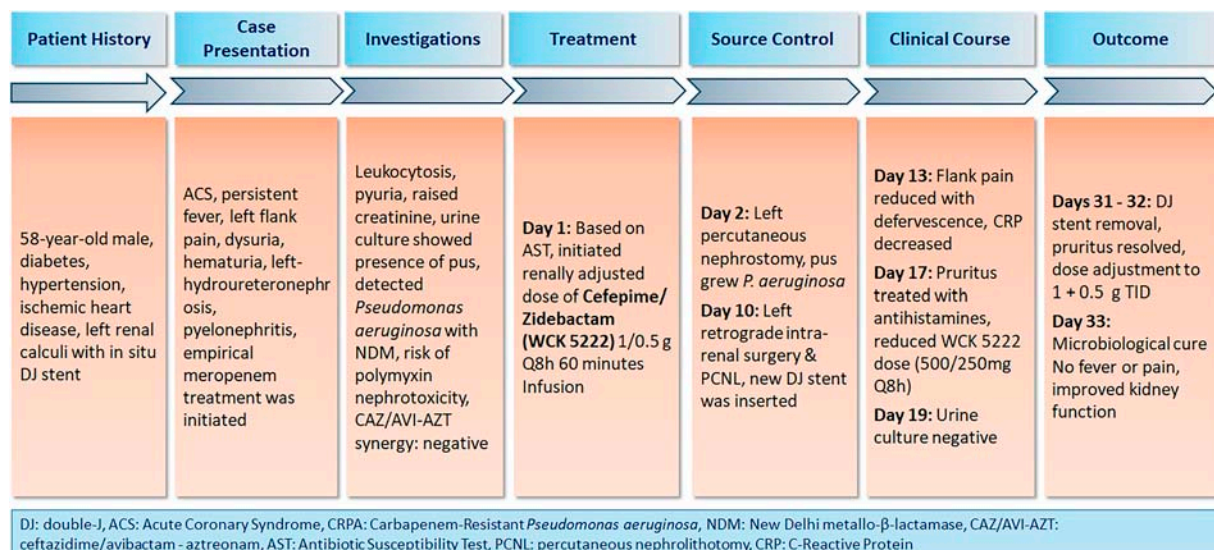


FIG. 2. Timeline of key clinical events, treatment progression, microbiological findings, and outcome.

This case exemplifies how otherwise treatable infections can become nearly untreatable due to resistance mechanisms such as NDM and demonstrates the potential utility of investigational β -lactam enhancer mechanism of action based, ceftipime/zidebactam. NDM is endemic in India,⁴ commonly observed in Enterobacterales, *P. aeruginosa*, and *Acinetobacter baumannii*, and the isolate in this case belonged to an internationally recognized high-risk clone with NDM,²² likely selected under prior meropenem exposure. This experience underscores the urgent need for novel agents such as ceftipime/zidebactam to manage high-risk resistant infections.

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Authors' Contributions

B.T.: Data curation and writing—original draft. S.M.: Conceptualization, data curation, formal analysis, investigation, supervision, visualization, writing—original draft, and writing—review and editing. D.P.: Data curation. P.N.: Data curation. B.S.: Supervision. K.S.: Data curation and Supervision. B.V.: Conceptualization, data curation, investigation, writing—original draft, and writing—review and editing. All authors reviewed the article.

Data Availability

This article does not report data generation or analysis.

Ethical Approval

Zidebactam/ceftipime treatment was initiated on compassionate grounds with due approval from The Central Drugs Standard Control Organisation (Indian Drug Regulator) and also from Ethics Committee of Care Institute of Medical Science (EC/NEW/INST/2023/GJ/0368). Written informed consent for the use of ceftipime/zidebactam was obtained from the patient's relative.

Disclosure Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Material

Supplementary Table S1

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